# Autosomal Recessive Inheritance

-Individuals heterozygous for the recessive gene are usually normal in phenotype. Thus the clinical disorder shows up only in the recessive homozygote.

## <u>**Cystic fibrosis:**</u>

-The affected individuals secrete thick mucus that clogs the air passages of their lungs, pancreas , obstructs pancreas ,and also affects  kideneys , liver and intestine.

- Because this mucus cannot be cleared effectively the lungs are highly susceptible to infection by bacteria such as ***Staphylococcus aureus*** and ***Pseudomonas aeruginosa***.

-Chronic obstruction and infection lead to destruction of the lung tissues, difficulty breathing and coughing up mucous, resulting eventually in death mostly by the time 4 or 5 years old if left untreated.

-The chloride pump channels that regulates the transport of $Cl^-$ ion a cross membrane is defective or absent in patients with CF.

-The sweat glands of CF patients are abnormal, resulting in high levels of chloride in the sweat.

-Chloride accumulates in cells which take up water by osmosis from the mucus that coats the cells.

-The prevalence among Caucasians is about 1\2500 but is much rarer in other races.

-Causes : Mostly due to mutation in CFTR gene /ΔF508

**Tay - Sachs disease: (lysosomal storage disease)**

Tay-sachs disease is rare in most human populations. However, it has a high incidence among Jews of Europe (Ashkenazim) and among American Jews. Approx.1 in 3600 of such Jewish infants has this disorder. Tay-sachs homozygotes appear normal at birth, but by about 8 months the symptoms become manifest. The infant begins to suffer from seizure, deafness,blindness, degeneration of motor and mental performance.

-The child dies with in a few years. Homozygous individuals lack an enzyme (Hexosaminidase A ) which is necessary to break down a special class of lipid called $G_{M2}$ ganglioside which occurs with in the lysosomes of brain cells. As a result the lysosomes fill with gangliosides swell and eventually burst, releasing oxidative enzymes that kill the brain cells.

**Albinism: (Oculocutaneous albinism and Ocular albinism)**

-An albino is an individual that lacks pigmentation in skin, hair and eyes. This is due to an inability to make the brown pigment melanin. Melanin is produced in pigment cells via different enzymatic pathways from the amino acid tyrosine. Most albino lack one of the enzymes necessary to produce melanin.

An albino displays

-Nystagmus (rapid uncontrolled eye movements).

-Strabismus(deviation of the eye from its normal axis).

-Reduced visual acuity, photophobia and red eyes.

Can two albinos have a normal child ?

## **Phenyl Ketonuria (PKU):(inborn errors of metabolism)**

Individual with PKU lacks the enzyme (phenylalanine hydroxylase) that normally converts the amino acid phenylalanine to tyrosine. When this enzyme is missing or deficient, phenylalanine and its abnormal break down products( phenyl pyruvate) accumulate in the blood-stream and urine.

These products are harmful to the cells of the developing nervous system and can result in profound mental retardation. Such infants usually appear healthy and normal at birth but after the first few months the symptoms of disease set in and with out treatment serve mental retardation usually results.

Patients display slow growth ,musty odor, and fair hair and skin. Many never learn to walk or talk and are subject to periodic convulsions. Affected individuals rarely live more than 30 years. PKU can be treated nutritionally those identified at birth are put on a special diet containing low amount of phenylalanine enough to supply dietary needs but not enough to permit toxic accumulations. As a result they are able to develop normally

## Autosomal dominant inheritance (disorders):

**Huntington disease:(Late onset disease)**

-A degenerative disease of the nervous system that caused by a lethal dominant allele that most people appear normal until they are of middle age (35-45 years old).

-This neurological disorder leads to progressive degeneration of brain cells, which in turn causes severe muscle spasm, loss of motor control (chorea) and psychiatric problems, including dementia and affective disorders.

-There is no effective treatment, but some drugs are used such as **benzodiazepines** to help control the choreic movements. Affective disturbances are sometimes controlled with antipsychotic drugs and tricyclic antidepressants. Death comes 10 to 15 years after the on set of symptoms.

## Neurofibromatosis type1(NF1):

Affecting approx. 1 in 3000 individuals. It offers variable expression in a genetic disease. Some patients have only a few *café-au-lait* spots(hyperpigmented skin patches), Lisch nodules(benign growth of iris) and perhaps neurofibromas (non malignant of the peripheral nerve tumors).

Other patients have a much more severe expression of the disorder, including hundreds to thousands of neurofibromas, optic gliomas( benign tumor of optic nerve), learning disabilities, hypertension, scoliosis(lateral curvature of the spine), and malagnancies.

A mutation in the *NF1* gene mapped to chromosome 17 encoding **neurofibromin** tumor suppressor results in this condition.

**<u>Achondroplasia (Dwarfs):</u>**

-Achondroplastic dwarfism has an incidence of one case among 10,000 people. Heterozygous individuals have the dwarf phenotype(dispropotionate short stature), big head, frontal bossing and low nasal root ,and redundant skin folds in the arms and legs but, homozygosity for this dominant allele is usually lethal, causing spontaneous abortion. Although achondroplastic dwarfs are less likely to have children than are other people there is a high rate of new mutation in the gene involved.

Cause: Mutation in fibroblast growth factor receptor3(FGR3) gene

# Achondroplasia features